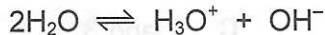


- 13 Use of the Data Booklet is relevant to this question.

Pure water ionises according to the equation shown.



The equilibrium constant for this reaction, K_c , is given by the expression shown.

$$K_c = \frac{[\text{H}_3\text{O}^+][\text{OH}^-]}{[\text{H}_2\text{O}]^2}$$

The density of pure water at 298 K is 0.997 g cm^{-3} .

Which expression gives the number of H_3O^+ ions that are present in 1.00 dm^3 of pure water at 298 K?

A $\frac{0.997}{18} L K_c$

B $\frac{0.997}{18} L \sqrt{K_c}$

C $\frac{997}{18} L K_c$

D $\frac{997}{18} L \sqrt{K_c}$